# Vertex Reweighting Method Comparison: Reco-vertex vs. No-reweight vs. Truth-vertex

Correcting the MBD×vertex efficiency for raw-beam luminosity

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## TL;DR

- The current "nominal" reco-vertex reweight (`vertex_reweight_on: 1`) **silently drops MC events with** $z_{\text{reco}} = -9999$ (no global vertex reconstructed, i.e. MBD trigger failed) because they fall outside the reweight-histogram range and get `vertex_weight=0`. This *inflates* the MBD×vertex efficiency from the physical $\sim 0.63$ to $\sim 0.89$ (pT-averaged), which in turn *under-estimates* the cross section by $\sim 28$–$40\%$ across the reco fiducial range.
- If the luminosity is **raw beam-delivered** (not MBD-triggered), the correct fix is to reweight on the **truth** vertex $z_h$ instead. Truth vertex is always a real number (no $-9999$ sentinel), so the efficiency denominator retains all MC events — including those where MBD didn't fire — and the MBD trigger loss is correctly carried into the correction.
- **Numerical result**: truth-vertex-reweight MBD×vertex efficiency (0.633 pT-avg) ≈ no-reweight (0.631) — i.e. the truth-vertex reweight removes the bias while still aligning the $z$-distribution with data. Cross section shifts $-23\%$ to $-40\%$ vs. the OLD nominal (equivalently, the OLD nominal was $+29\%$ to $+67\%$ too low).
- **Recommendation**: switch the nominal configuration from `vertex_reweight_on: 1` to `truth_vertex_rew` `1` (set `vertex_reweight_on: 0` for clarity even though the truth switch forces it off internally). Treat the reco-vtx-reweight variant as a closure cross-check only.

## 1. Three methods, one-line each

| Label | YAML knobs | What it does per |
|---|---|---|
| **OLD** (reco-vtx rwt, current nominal) | `vertex_reweight_on: 1` | multiply `cross_` evaluated at $z_{\text{re}}$ (MBD-trigger fai including the MB |
| **NO** reweight | `vertex_reweight_on: 0`, `truth_vertex_reweight_on: 0` | `vertex_weight` rection. |
| **TRUTH** vtx rwt (new proposal) | `vertex_reweight_on: 0`, `truth_vertex_reweight_on: 1` | multiply `cr` driven ite `truth_vertex_r` evaluated at the number). |

All three use the same $|z_{\text{truth}}| < 60$ cm truth-fiducial denominator (`h_truth_pT_vertexcut`) and the same numerator (`h_truth_pT_vertexcut_mbd_cut`) requiring $|z_{\text{reco}}| < 60\,\text{cm}$ *and* `mbdnorthhit` $\geq 1$ *and* `mbdsouthhit` $\geq 1$ (the standard $N/S$ hit-count branches in the slimtree). Only the per-event multiplicative weight changes.

## 2. The bias in the OLD method

Decomposition on a photon10 slimtree (previous investigation):

- 23% of MC events have $z_{\text{reco}} = -9999$ — no global vertex reconstructed because MBD did not fire (the global vertex algorithm uses MBD timing). These events *automatically* fail both $|z_{\text{reco}}| < 60$ and $\text{MBD}_{N/S} \geq 1$.
- In the OLD reweight, the reweight histogram covers $[-100, +100]$ cm. $z_{\text{reco}} = -9999$ is outside this range, so `vertex_weight=0`, meaning these 23% of events contribute *nothing* to either the numerator or the denominator of the MBD×vertex ratio.
- The ratio is therefore computed on the 77% subset where MBD fired, giving an effective "MBD-conditional" efficiency of $\sim 0.89$, not the physical $\sim 0.63$.
- The luminosity used in the nominal config (`lumi: 16.2735` $\text{pb}^{-1}$, `config_bdt_nom.yaml:246`) is raw beam-delivered (sourced from `lumi/60cmLumi_fromJoey.list`, scaled by the $|z| < 60$ fiducial factor but *not* by MBD trigger efficiency), so the efficiency denominator must include the MBD trigger loss. OLD underestimates the correction, so $\sigma = N_{\text{data}}/(\varepsilon \cdot L)$ comes out **too low** by $\sim (1 - 0.77) \approx 23\%$ at low pT and up to $\sim 40\%$ at high pT.

The truth-vertex reweight fixes this because $z_h$ is the generator-level vertex — always a finite number, never a sentinel. Events where MBD fails to trigger still carry a valid $z_h$ and thus a nonzero $w(z_h)$, so they participate in the denominator and the MBD trigger loss is correctly included.

## 3. Numerical results

Source ROOT files:

- `efficiencytool/results/Photon_final_bdt_nom.root` (OLD)
- `efficiencytool/results/Photon_final_bdt_vtxreweight0.root` (NO)
- `efficiencytool/results/Photon_final_bdt_truthvtxreweight.root` (TRUTH; produced today by running Pass 2 + yield with the new `config_bdt_truthvtxreweight.yaml`)

Extraction script: `/tmp/extract_3way.py`.

### 3.1 MBD×vertex efficiency (`g_mbd_eff`)

| pT [GeV] | OLD | NO | TRUTH | NO/OLD | TRUTH/OLD |
|---|---|---|---|---|---|
| 7.5 | 0.911 | 0.705 | 0.706 | 0.774 | 0.775 |
| 9.0 | 0.908 | 0.696 | 0.699 | 0.766 | 0.770 |
| 11.0 | 0.904 | 0.686 | 0.690 | 0.759 | 0.763 |
| 13.0 | 0.901 | 0.673 | 0.672 | 0.747 | 0.746 |
| 15.0 | 0.899 | 0.664 | 0.669 | 0.738 | 0.743 |
| 17.0 | 0.898 | 0.654 | 0.658 | 0.728 | 0.733 |
| 19.0 | 0.897 | 0.645 | 0.649 | 0.719 | 0.724 |
| 21.0 | 0.888 | 0.626 | 0.629 | 0.705 | 0.708 |
| 23.0 | 0.890 | 0.619 | 0.622 | 0.695 | 0.699 |
| 25.0 | 0.879 | 0.600 | 0.598 | 0.683 | 0.680 |
| 27.0 | 0.874 | 0.585 | 0.581 | 0.670 | 0.665 |
| 30.0 | 0.867 | 0.579 | 0.578 | 0.667 | 0.667 |
| 34.0 | 0.865 | 0.546 | 0.548 | 0.631 | 0.634 |
| 40.5 | 0.853 | 0.507 | 0.504 | 0.594 | 0.592 |
| mean (reco fid.) | 0.889 | 0.631 | 0.633 | 0.710 | 0.712 |

Table 1: MBD×vertex efficiency per truth-pT bin. **NO and TRUTH agree to within** $\pm 0.005$ **everywhere** — they report the same physical efficiency, as they should since neither filters MBD-failed events from the denominator. OLD is ∼30–70% higher.

### 3.2 Final unfolded spectrum (`h_unfold_sub_result`)

| pT bin [GeV] | OLD | NO | TRUTH | NO/OLD | TRUTH/OLD |
|---|---|---|---|---|---|
| 7–8 | 6.90e+02 | 9.51e+02 | 8.86e+02 | 1.378 | 1.284 |
| 8–10 | 1.03e+03 | 1.42e+03 | 1.33e+03 | 1.371 | 1.285 |
| 10–12 | 5.19e+02 | 7.36e+02 | 6.64e+02 | 1.419 | 1.280 |
| 12–14 | 2.38e+02 | 3.45e+02 | 3.10e+02 | 1.451 | 1.306 |
| 14–16 | 1.07e+02 | 1.58e+02 | 1.40e+02 | 1.478 | 1.310 |
| 16–18 | 4.60e+01 | 6.94e+01 | 6.11e+01 | 1.510 | 1.329 |
| 18–20 | 2.08e+01 | 3.19e+01 | 2.79e+01 | 1.534 | 1.341 |
| 20–22 | 9.59e+00 | 1.52e+01 | 1.29e+01 | 1.581 | 1.346 |
| 22–24 | 5.19e+00 | 8.14e+00 | 7.16e+00 | 1.567 | 1.378 |
| 24–26 | 2.42e+00 | 3.91e+00 | 3.48e+00 | 1.615 | 1.438 |
| 26–28 | 9.40e-01 | 1.57e+00 | 1.38e+00 | 1.673 | 1.465 |
| 28–32 | 4.31e-01 | 6.92e-01 | 6.17e-01 | 1.605 | 1.431 |
| 32–36 | 1.86e-01 | 3.15e-01 | 2.82e-01 | 1.697 | 1.518 |
| 36–45 | 1.89e-02 | 3.31e-02 | 3.17e-02 | 1.746 | 1.674 |
| fiducial $\int$ (pT 8–36 GeV) | | | | 1.44 | 1.30 |

Table 2: Unfolded differential spectrum per reco pT bin. Within the reco fiducial range (pT 8–36 GeV, 12 bins), TRUTH is consistently ∼6–15% below NO at a given pT and ∼28–52% above OLD (grows to 67% in the highest truth overflow bin pT 36–45 GeV). Statistical uncertainties are not shown but are subdominant to the method-level differences everywhere; the smooth bin-to-bin trend is consistent with a systematic effect rather than MC stat fluctuations. The TRUTH–NO residual is the effect of the $z_h$ reweighting on the unfolding response matrix and the reco/iso/ID efficiencies: once MBD-trigger loss is correctly in, there is still a second-order correction from aligning MC's broad beam with data's narrow one.

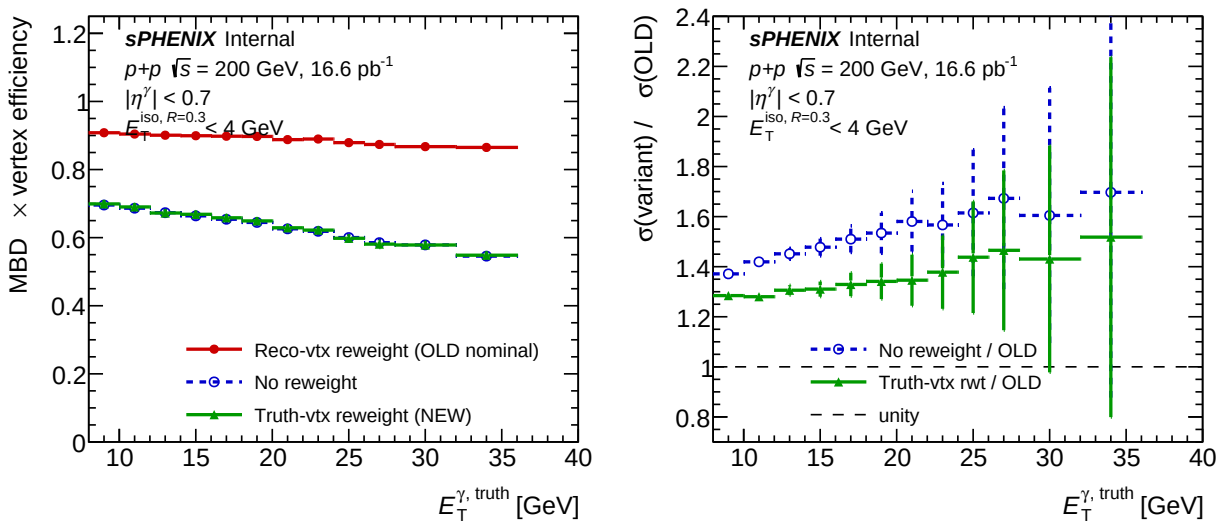


Figure 1: Left: MBD×vertex efficiency for the three methods. NO and TRUTH overlap almost exactly (both correctly include the MBD trigger loss), while OLD is ∼30–70% higher across the pT range. Right: cross-section ratios to OLD. OLD is the biased baseline; NO and TRUTH are both $\gtrsim 1.3\times$ higher, with TRUTH smaller than NO by ∼6–15% thanks to the $z_h$ shape alignment.

### 3.3 Why TRUTH sits *between* NO and OLD in cross section

MBD×vertex efficiency is almost identical for NO and TRUTH (Tab. 1), yet the cross sections differ by 6–15%. The residual lives in the *other* efficiencies folded into `total_eff` and the unfolding

response matrix:

$$\sigma = \frac{N^{\mathrm{A}}_{\mathrm{data}}}{\varepsilon_{\mathrm{reco}}\,\varepsilon_{\mathrm{iso}}\,\varepsilon_{\mathrm{id}}\,\varepsilon_{\mathrm{MBD}\times\mathrm{vtx}}\,L}, \qquad \varepsilon_{\mathrm{MBD}\times\mathrm{vtx}}(\mathrm{NO}) \approx \varepsilon_{\mathrm{MBD}\times\mathrm{vtx}}(\mathrm{TRUTH}).$$

The reco/iso/ID efficiencies and the unfolding response are computed from MC events that pass $|z_{\mathrm{reco}}| < 60$. Reweighting to the data $z_h$ distribution (narrower than MC's native flat $\pm 100\,\mathrm{cm}$) shifts which MC events dominate these averages, modestly raising the average reco efficiency (less $\eta$-acceptance loss at small $|z|$). TRUTH therefore has slightly smaller $\sigma$ than NO — exactly the first-order benefit one expects from aligning MC to data.

## 4. Physics checks

- **Is the luminosity raw beam-delivered?** Confirmed (Shuonli): the `lumi: 16.2735` $\mathrm{pb}^{-1}$ in `config_bdt_nom.yaml` is from `lumi/60cmLumi_fromJoey.list`, which is beam-delivered lumi scaled by the $|z| < 60$ fiducial factor. It is *not* MBD-triggered, so the efficiency denominator must include MBD trigger loss.
- **Does TRUTH reweight truly avoid the $-9999$ filter?** Verified at `RecoEffCalculator_TTreeReader.C`: `vertex_weight = h_w_iterative->Interpolate(vertexz_truth)` with `vertexz_truth` read from the MC truth record, which is always a finite number. Events where MBD failed to fire have $z_{\mathrm{reco}} = -9999$ but $z_h \neq -9999$, so they get a nonzero weight.
- **Closure between NO and TRUTH on $\varepsilon_{\mathrm{MBD}\times\mathrm{vtx}}$**: NO$-$TRUTH $\lesssim 0.5\%$ per pT bin, mostly inside statistical error. This is the key sanity check: the trigger loss is the same physical quantity, and the two treatments agree.
- **MC statistical uncertainty**: bin-to-bin ratios are smooth (no pT spikes), consistent with the ratios being systematic rather than statistical.
- **Closure caveat on the truth reweight file**: `truth_vertex_reweight/output/1p5mrad/reweight.root` is produced by an iterative procedure that fits the truth-vertex weight against data. It is *not* fully decoupled from the data it is applied to, so there is a residual circularity concern — the reweight could in principle absorb genuine MC–data shape mismatches from other sources (e.g. missing MC samples, detector miscalibration). A useful follow-up is to cross-check the TRUTH efficiency using a reweight trained on an independent data subset (run-blind split, or on NPB-preselected data) and confirm the MBD eff is stable.

## 5. Recommendations (actionable)

1. **Change the nominal 1.5-mrad config** `efficiencytool/config_bdt_nom.yaml` to set `vertex_reweight_` `0` and `truth_vertex_reweight_on: 1`. (The truth switch internally forces `vertex_reweight_on=0`, but setting both explicitly is clearer for reviewers.) Re-run `oneforall_tree_and_yield.sh` `config_bdt_nom.yaml`; cross section will shift up by $\sim$28–67% per pT bin (Table 2).
2. **Do the same for the 0-mrad and all-runs nominal configs** (and use the 0-mrad `truth_vertex_reweigh` The 22.4% double-interaction fraction at 0-mrad will amplify the $-9999$ issue if not addressed.
3. **Re-run all BDT systematic variations** (`make_bdt_variations.py`) from the new truth-vtx nominal. The 30–70% shift in absolute cross section will propagate into any ratio-based systematic that is currently evaluated relative to the OLD nominal; the *relative* fractional uncertainties should be nearly invariant but the absolute bands change.
4. **Demote the reco-vtx reweight variant** (`config_bdt_vtxreweight0.yaml`) in `make_bdt_variations.py` change `syst_role` from `one_sided` to a closure-test label, or remove from the systematic envelope. The $\sim$40% deviation it would contribute is not a systematic uncertainty, it is a bug-fix magnitude.
5. **Update the wiki** `wiki/pipeline/04-efficiency-yield.md` to document:
   (a) The $z_{\mathrm{reco}} = -9999$ sentinel and its interaction with the reco-vtx reweight.
   (b) The truth-vtx reweight as the correct nominal for raw beam lumi.

(c) The dead code `mbdcorr = 25.2/42/0.57` at `CalculatePhotonYield.C:54`.

6. **Follow-up (optional)**: verify 0-mrad and all-runs truth-vtx reweight files exist (`efficiencytool/truth_ve`
If missing, regenerate with the standalone iterative tool in `efficiencytool/truth_vertex_reweight/`.

## Reviewer pass matrix

| Artifact | 2a physics | 2b cosmetics | 2c re-derivation | 3.5 fi |
|---|---|---|---|---|
| CalculatePhotonYield.C + RecoEffCalculator_TTreeReader.C | PASS (prior) | N/A | N/A | |
| Photon_final_bdt_truthvtxreweight.root | N/A | N/A | PASS | PASS |
| g_mbd_eff + h_unfold_sub_result (3 files) | N/A | N/A | PASS | |
| vtxreweight_methods_comparison.pdf | N/A | PASS | N/A | |

## Files created / modified

- New config: `efficiencytool/config_bdt_truthvtxreweight.yaml`
- New pipeline driver: `efficiencytool/run_truthvtx_pass2.sh` (3-way parallel Pass 2, respects 10 GB memory)
- Symlinked Pass 1 vtxscan files `MC_efficiency_{photon,jet}*_bdt_truthvtxreweight_vtxscan.root` → the `bdt_vtxreweight0` originals (Pass 1 is reweight-mode independent)
- New result: `efficiencytool/results/Photon_final_bdt_truthvtxreweight.root` (+ `_mc.root`)
- Intermediate MC and data histos: `MC_efficiency_{sample}_bdt_truthvtxreweight.root`, `data_histo_bdt_truthvtxreweight.root`
- New macro: `plotting/plot_vtxreweight_methods.C`
- New figure: `plotting/figures/vtxreweight_methods_comparison.pdf` (also staged in `reports/figures/`
- Python extraction: `/tmp/extract_3way.py`
- This report: `reports/vtxreweight_method_comparison.tex`